

Warm-up: functions that we hand over

Programming Paradigms

Session 5 — the warm-up, before the quest

Part 1 goes with block 1 of the lecture, Part 2 with block 2.

In each part: questions on paper, then exercises at the keyboard, with `warmup.ml`.

SOLUTIONS

In the lecture a function was handed over twice: as one step of a traversal, and as a computation not yet run. Before the rope and the editor, we try both on things everyone has met: a list of numbers, a counter, a triangle of numbers, and the queue of lab 2. Nothing here is themed. The notation is OCaml's, and this is all we need:

<code>fun x -> e</code>	the function that takes <code>x</code> and gives <code>e</code> ; it has no name
<code>let f x y = e</code>	the same as <code>let f = fun x -> fun y -> e</code>
<code>f x y</code>	<code>f</code> applied to <code>x</code> , and the result applied to <code>y</code>
<code>fun () -> e</code>	a function that waits for nothing but its call: a thunk
<code>ref 0, !n, n := 1, incr n</code>	a cell that can change; read it; write it; add one
<code>lazy e, Lazy.force l</code>	a value computed at the first <code>force</code> , and kept

The file `warmup/warmup.ml` holds the exercises at the keyboard, with holes marked `TODO` and several checks for each. The exercises are stated here; the file only says where to type. It needs no dune:

```
ocaml warmup/warmup.ml          # pass / FAIL / todo, one line per check
```

Part 1 — functions, folds, iterators

On paper

Question A (a function is a value)

Given the three definitions, write the value of each of the five expressions under them, and the types of `add`, of `add3` and of `twice`.

```
let twice f x = f (f x)
let add n = fun x -> x + n
let add3 = add 3
```

```
add3 10
twice add3 10
twice (add 1) 0
List.map (add 10) [1; 2; 3]
twice twice add3 10
```

Then, given the function `add_pair`, which takes its two numbers as one pair, write `add3` from it. Can we do it without writing a new function?

```
let add_pair (n, x) = x + n
```

Answer. 13; 16; 2; [11; 12; 13]; 22. In the last one, `twice twice` is `twice` applied to itself: it applies its argument four times, and four times 3 added to 10 is 22.

```
add    : int -> int -> int
add3   : int -> int
twice  : ('a -> 'a) -> 'a -> 'a
```

From `add_pair` we must write a new function:

```
let add3 = fun x -> add_pair (3, x)
```

`add_pair` wants its two numbers at once, and we have one. `add` takes them one after the other, so `add 3` is already the function we want: `add` is curried, and `add 3` is a partial application.

Question B (what a function carries)

Given the function `counter` and the five definitions under it, write the values of `x`, `y` and `z`.

```
let counter () =  
  let n = ref 0 in  
  fun () -> incr n; !n  
  
let a = counter ()  
let b = counter ()  
let x = a ()  
let y = a ()  
let z = b ()
```

Then, given the list of functions `fs`, write the value of the last expression.

```
let fs = List.map (fun i -> fun () -> i * 10) [1; 2; 3]  
  
List.map (fun f -> f ()) fs
```

Answer. `x` is 1, `y` is 2, `z` is 1. Every call of `counter` makes a new cell `n`; `a` and `b` each carry their own, and no other code can reach it. The last expression is `[10; 20; 30]`: each of the three functions carries its own `i`, which never changes. (In JavaScript, with `var`, the three functions of the lecture shared one variable.)

At the keyboard

Exercise 1 (functions that take and return functions)

Given `add` and `add_pair`, write `add3` from `add`, and `add3'` from `add_pair`. Then write `twice`, which applies a function two times, and `compose`, which applies `g` and then `f`.

```
add3      : int -> int  
add3'     : int -> int  
twice     : ('a -> 'a) -> 'a -> 'a  
compose   : ('b -> 'c) -> ('a -> 'b) -> 'a -> 'c
```

Answer.

```
let add3 = add 3  
let add3' = fun x -> add_pair (3, x)  
let twice f x = f (f x)  
let compose f g = fun x -> f (g x)
```

Exercise 2 (a closure with a state of its own)

Given `counter`, write `accumulator`: each call of `accumulator ()` makes a function with a total of its own, which starts at 0. That function adds its argument to the total and returns the total.

```
accumulator : unit -> int -> int
```

Answer.

```
let accumulator () =  
  let total = ref 0 in  
  fun x ->  
    total := !total + x;  
    !total
```

The `let total` is before the `fun`: it is run once for each call of `accumulator ()`. Written after the `fun`, the total would start again at 0 at every call of the function.

Exercise 3 (three functions, written with a fold)

Given `fold`, write the sum of a list of numbers, the number of elements that satisfy a predicate, and the list in the opposite order. Each is one call of `fold` and has no recursion of its own.

```
let rec fold f acc l =
  match l with
  | [] -> acc
  | x :: xs -> fold f (f acc x) xs

sum    : int list -> int
count  : ('a -> bool) -> 'a list -> int
rev    : 'a list -> 'a list
```

Answer.

```
let sum l = fold (fun acc x -> acc + x) 0 l
let count p l = fold (fun n x -> if p x then n + 1 else n) 0 l
let rev l = fold (fun acc x -> x :: acc) [] l
```

All three need every element of the list: none has a reason to stop before the end.

Exercise 4 (a fold that can stop)

Given `fold_until`, whose function answers `Continue` to go on and `Stop` to end the fold, write `find_first`, the first element that satisfies a predicate, and `sum_until_negative`, the sum of the numbers before the first negative one. Each is one call of `fold_until`.

```
type 'b go =
  | Continue of 'b
  | Stop of 'b

let rec fold_until f acc l =
  match l with
  | [] -> acc
  | x :: xs ->
    match f acc x with
    | Stop a -> a
    | Continue a -> fold_until f a xs

find_first      : ('a -> bool) -> 'a list -> 'a option
sum_until_negative : int list -> int
```

One check counts the calls of the predicate: on `[3; 12; 7; 40]`, looking for a number over 10, there are two.

Answer.

```
let find_first p l =
  fold_until
    (fun _ x -> if p x then Stop (Some x) else Continue None)
    None l

let sum_until_negative l =
  fold_until
    (fun acc x -> if x < 0 then Stop acc else Continue (acc + x))
    0 l
```

With `fold`, `find_first` would give the same answer, and still visit the four elements.

Exercise 5 (an iterator, as a closure)

Given two numbers `a` and `b`, write `range`, the iterator that gives `a`, then `a + 1`, up to `b - 1`, one number at each call, and `None` at every call after that. Then write `drain`, the list of the elements that an iterator still has to give.

```
range : int -> int -> unit -> int option
drain : (unit -> 'a option) -> 'a list
```

Answer.

```
let range a b =
  let next = ref a in
  fun () ->
    if !next >= b then None
    else begin
      let x = !next in
      next := x + 1;
      Some x
    end

let rec drain next =
  match next () with
  | None -> []
  | Some x -> x :: drain next
```

`range` is the counter again, with a place in the place of a count. `drain` changes the iterator it reads: called a second time on the same one, it returns the empty list.

Part 2 — thunks, sequences, lazy

On paper

Question C (when is it printed?)

Given the program, write what it prints, character by character, in order.

```
let t = fun () -> print_string "A"; 1
let l = lazy (print_string "B"; 2)

let () = print_string "C"

let x = t () + t ()
let y = Lazy.force l + Lazy.force l

let () = Printf.printf " %d %d" x y
```

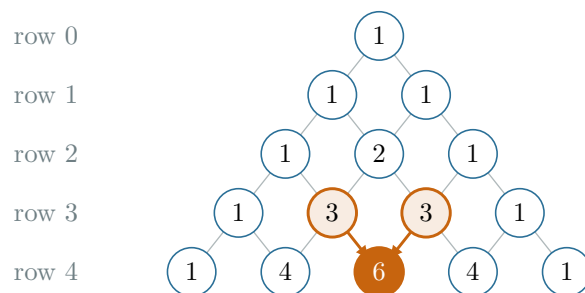
Answer. CAAB 2 4. Making the thunk and the lazy value prints nothing, so C comes first. The thunk runs its body at each of its two calls: AA. The lazy value runs its body at the first `force` and keeps the 2: one B.

Question D (a triangle of numbers)

In Pascal's triangle every row begins and ends with 1, and every other number is the sum of the two numbers above it. We write $T(n, k)$ for the number of row n at column k , where the rows and the columns are counted from 0. The triangle is defined by recursion on n :

$$\begin{aligned} T(n, 0) &= 1 \\ T(n, n) &= 1 \\ T(n, k) &= T(n-1, k-1) + T(n-1, k) \quad \text{for } 0 < k < n \end{aligned}$$

In the drawing, $T(4, 2) = T(3, 1) + T(3, 2)$ is marked: $6 = 3 + 3$.



The same definition, read one row at a time: we write row $n - 1$ twice, the first time with a 0 before it and the second time with a 0 after it, and we add the two column by column. The result is row n .

$$\begin{array}{cccccc}
 & 0 & 1 & 3 & 3 & 1 \\
 + & 1 & 3 & 3 & 1 & 0 \\
 = & 1 & 4 & 6 & 4 & 1
 \end{array}$$

Given the row 1 4 6 4 1, write the two rows that follow. Then say, for the first line of the addition and for the second one, which number of the definition stands at column k , and where the two 1 at the ends of a row come from. Last, write the sum of each of the rows 0 to 5. What are these sums?

Answer. 1 5 10 10 5 1, then 1 6 15 20 15 6 1. With the 0 before it, row $n - 1$ is shifted by one column to the right: at column k the first line has $T(n - 1, k - 1)$. The second line has $T(n - 1, k)$. Their sum is $T(n, k)$, the third equation. At column 0 the first line has the 0 and the second one has $T(n - 1, 0) = 1$; at column n the second line has the 0 and the first one has $T(n - 1, n - 1) = 1$: these are the first two equations. The sums are 1, 2, 4, 8, 16, 32: each row is added to itself, shifted, so its sum is twice the sum of the row above.

At the keyboard

The type of sequences is the lecture's, and `take` is given.

```

type 'a node =
  | Nil
  | Cons of 'a * 'a seq

and 'a seq = unit -> 'a node

```

Exercise 6 (a row of the triangle, from the row above it)

Given two lists, write `zip_with`, which combines them element by element with a function and stops at the end of the shorter one. Then write `next_row`, by the rule of question D, as one call of `zip_with`.

```

zip_with : ('a -> 'b -> 'c) -> 'a list -> 'b list -> 'c list
next_row : int list -> int list

```

From the definition to the code. If `row` is row $n - 1$, the list `0 :: row` has $T(n - 1, k - 1)$ at position k , and the list `row @ [0]` has $T(n - 1, k)$ there. The three equations of the definition become one addition of two lists: `next_row` has no if and no case for the ends of the row.

Answer.

```

let rec zip_with f l1 l2 =
  match l1, l2 with
  | x :: xs, y :: ys -> f x y :: zip_with f xs ys
  | _ -> []

let next_row row = zip_with ( + ) (0 :: row) (row @ [0])

```

Exercise 7 (a sequence without end)

Given a function `f` and a value `x`, write `iterate`, the sequence `x, f x, f (f x)`, and so on, without end. The line that counts the cells built is given; only the cell is to write.

```
iterate : ('a -> 'a) -> 'a -> 'a seq
```

One check makes the sequence and reads the counter before anything is asked for: it is 0. Another counts the calls of `f`: to read five elements, `f` is called four times, and not five.

From the definition to the code. A definition by recursion on n has a first value and a step from $n - 1$ to n . `iterate` receives exactly these two: `x` is the first value and `f` is the step.

Answer.

```
let rec iterate f x =  
  fun () ->  
    incr built;  
    Cons (x, fun () -> iterate f (f x) ())
```

The recursive call is under `fun () ->`: it is not run before the sequence is called. With a list, `x :: iterate f (f x)` would never return.

The rest has a `fun () ->` of its own. With `Cons (x, iterate f (f x))` the sequence is right, and one step ahead: `f x` is an argument, and OCaml computes an argument before the call (session 4), so the cell of `x` would compute `f x` when it is built, and five elements would cost five calls of `f`. Under its own `fun () ->`, `f x` is computed when the rest is asked for. The library's `Seq.iterate` is written in this way.

Exercise 8 (the triangle, as the sequence of its rows)

Given `iterate` and `next_row`, write `pascal`, the sequence of all the rows of the triangle, and `row`, which returns row `n`, the top row being row 0.

```
pascal : int list seq  
row      : int -> int list
```

From the definition to the code. Row 0 is the list with the single number 1, and `next_row` is the step from row $n - 1$ to row n . The definition says how to compute any row; it does not say how many. With a sequence we do not have to say it either.

One check counts the rows built by `row 6`: there are seven, rows 0 to 6, and no more. When `pascal` is written, the file draws the triangle on the screen, two times: with its numbers, and with a star for every odd number.

Answer.

```
let pascal = iterate next_row [1]  
  
let row n = List.nth (take (n + 1) pascal) n
```

The triangle has no last row, and `pascal` is a value all the same. We compute as many rows as we ask for.

Exercise 9 (the diagonals of the triangle)

Given the rows of the triangle, write the one line that is missing in `diagonal`. Diagonal `n` starts in row `n` at column 0 and goes up to the right: row `n - 1` at column 1, row `n - 2` at column 2, as long as the column exists in the row. `diagonal n` is the sum of these numbers:

$$T(n, 0) + T(n - 1, 1) + T(n - 2, 2) + \dots$$

In the code, `k` is the column and `acc` the sum so far; `T(n - k, k)` is element `k` of element `n - k` of the list `rows`.

```
diagonal : int -> int
```

Then write the first ten sums on paper. Where have we seen these numbers?

Answer.

```

let diagonal n =
  let rows = take (n + 1) pascal in
  let rec go k acc =
    if 2 * k > n then acc
    else go (k + 1) (acc + List.nth (List.nth rows (n - k)) k)
  in
  go 0 0

```

1, 1, 2, 3, 5, 8, 13, 21, 34, 55: the Fibonacci numbers, which the lecture defined as a list that reads itself at two places. The lecture's list starts at 0: it is 0, 1, 1, 2, 3, ..., and the sum of diagonal n is its number $n + 1$.

Exercise 10 (a row computed once)

Given `row`, write `lazy_rows`: the list of n lazy values, where number i holds row i , not yet computed.

```

lazy_rows : int -> int list Lazy.t list

```

The checks make ten lazy rows and read the counter: no row is built. Then they read row 5 two times: six rows are built at the first reading, and none at the second.

From the definition to the code. The definition can also be written as it stands, as a function `t n k` with two recursive calls. That function computes the same numbers again and again, as the list of Fibonacci numbers did with thunks: $T(3, 1)$ is needed by $T(4, 1)$ and by $T(4, 2)$. Computed by rows, a row is computed once for all the numbers of the row under it; kept in a lazy value, it is not computed again when it is asked for again.

Answer.

```

let lazy_rows n = List.init n (fun i -> lazy (row i))

```

With `fun () -> row i` in the place of `lazy (row i)`, the second reading would build the six rows again: a thunk delays, a lazy value delays and remembers.

Each lazy value remembers its own row, and nothing more: reading row 6 after row 5 builds rows 0 to 5 again, since `row 6` goes through `pascal` from its start. To share the rows between the readers, the rest of the sequence itself must be a lazy value, as in the lecture's `fibs`.

Homework — lab 2's queue, with its reversal kept

This part is not for the lab session. It goes with the end of block 2, and it answers the question on which lab 2 ended.

On paper

Question E (a queue used twice)

Given lab 2's queue, a front and a rear, where `pop` reverses the rear when the front is empty, and the queue `q`, write the values of `a` and of `b`. How many times is the list `[3; 2; 1]` reversed?

```

let pop q =
  match q with
  | x :: front, rear -> Some (x, (front, rear))
  | [], rear ->
    match reverse rear with
    | [] -> None
    | x :: front -> Some (x, (front, []))

let q = ([], [3; 2; 1])

let a = pop q
let b = pop q

```

Then, given that the reversal is kept in the queue as a lazy value, which `a` and `b` both read, write how many times it is run.

Answer. `a` and `b` are both `Some (1, ([2; 3], []))`. The list is reversed two times: `q` is not changed by the first `pop`, so the second one finds the same empty front and does the same work. In lab 2 the cost of the queue was one reversal for the elements pushed, as long as each version was used once; here a version is used twice, and that count no longer holds. With a lazy value the reversal is run once: the first `force` runs it and keeps the list, the second one reads the list.

At the keyboard

Exercise 11 (lab 2's queue, with its reversal kept)

Given the type of the second queue, where the front is a lazy value and the two lengths are kept, write the two expressions that are missing. In `balance`: the new front, which is the old front followed by the rear reversed, and which must not be computed now. In `lpop`: the queue without its first element.

```
type 'a lqueue = {
  front : 'a list Lazy.t;
  flen : int;
  rear : 'a list;
  rlen : int;
}

balance : 'a lqueue -> 'a lqueue
lpush   : 'a -> 'a lqueue -> 'a lqueue
lpop    : 'a lqueue -> ('a * 'a lqueue) option
```

The file is `warmup/queue.ml`, and it is run as the other one. Its first check is question E, on lab 2's queue as it was. Its last check pops the same lazy queue two times: the second time, no reversal is run.

Answer.

```
let balance q =
  if q.rlen <= q.flen then q
  else
    let front = q.front and rear = q.rear in
    { front = lazy (Lazy.force front @ reverse rear);
      flen = q.flen + q.rlen;
      rear = [];
      rlen = 0 }

let lpop q =
  match Lazy.force q.front with
  | [] -> None
  | x :: rest ->
    Some (x, balance { q with front = lazy rest; flen = q.flen - 1 })
```

The reversal is put in the queue when the rear becomes longer than the front, as a lazy value, and it is run when the front is read. Two versions made from the same queue hold the same lazy value: whichever reads it first pays for the reversal, and the other finds the list.

This queue shows one reversal shared by two versions, and no more than that. It does not have the cost of lab 2's Part D when a version is used more than once: two pushes on the same queue make two lazy values, and each reversal is paid for. That cost is the banker's queue's (Okasaki, *Purely Functional Data Structures*, chapter 6), where the front is a sequence whose rest is a lazy value, read one cell at a time.